

MICROCANONICAL EXCESS-PRESSURE SEMIGROUPS AND KINETIC THETA-GENERATORS

QIAN QI

ABSTRACT. We distinguish the deterministic flow, Koopman observables, probability laws, factorial hierarchies, and limiting kinetic values. The dynamic entropy action has compact sublevels and defines a strongly continuous Nisio semigroup. Its discounted variational resolvent satisfies the nonlinear resolvent identity, has full range, and generates an m -dissipative graph with comparison. State-dependent controls are transferred by their relative multiplier rather than by reusing one collision measure. A single diagonal BBGKY graph-core corrector verifies all nonlinear Trotter–Kato hypotheses.

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1. ROUND-TWENTY-ONE POSITIVE CLOSURE: NONLINEAR RESOLVENTS,
REALIZABLE HIERARCHY CORES, AND SEMIGROUP CONVERGENCE

This section replaces the former Round-Seventeen source. The state space, the controlled action, and the function space are fixed before the semigroup is constructed. The discounted variational operator satisfies the nonlinear resolvent identity for an accretive graph; the linear pseudo-resolvent formula is nowhere used.

1.1. The kinetic state and compact action topology. Let $\mathcal{E}_M$ be the set of nonnegative densities f on $\mathbb{T}^3 \times \mathbb{R}^3$ with prescribed mass and with

$$\int (1 + |v|^2) f \, dx dv \leq M.$$

Choose a dense sequence $(\phi_j)_{j \geq 1}$ in the unit ball of smooth tests satisfying $|\phi_j| + |\nabla_x \phi_j| + |\Delta \phi_j| \leq C_j(1 + |v|^2)$, and put

$$d_*(f, g) = \sum_{j \geq 1} 2^{-j} \frac{|\int \phi_j(f - g)|}{1 + |\int \phi_j(f - g)|}.$$

The full state space is the increasing union $\mathcal{E} = \bigcup_M \mathcal{E}_M$ with weight $\mathfrak{w}(f) = 1 + \int (1 + |v|^2) f$. Let $\text{BUC}_{\mathfrak{w}}(\mathcal{E})$ be the functions h for which $h/\mathfrak{w}$ is bounded and uniformly continuous on every energy shell, with norm

$$\|h\|_{\mathfrak{w}} = \sup_f \frac{|h(f)|}{\mathfrak{w}(f)}.$$

For an admissible path (f, Γ) on $[0, T]$, write $\Gamma = qA_f$ and use the B2 action

$$\mathcal{A}_T(f, \Gamma) = \int_0^T \int \ell(q) \, dA_f, \quad \ell(q) = q \log q - q + 1.$$

The path satisfies the exact weak balance equation and begins from the stated initial condition.

Theorem 1.1 (Compact dynamic action sublevels). *Fix an energy shell $\mathcal{E}_M$, a time T , and $L < \infty$. The set of admissible paths beginning in $\mathcal{E}_M$ and satisfying $\mathcal{A}_T \leq L$ is compact after passing to the topology of uniform convergence of all density coordinates $t \mapsto \int \phi_j f_t$ and weak convergence of the contact current on compact time-velocity sets. The terminal-state map is continuous on this set, and the action is lower semicontinuous.*

Proof. The entropy inequality gives, for every nonnegative contact test G ,

$$\int G q \, dA_f \leq \mathcal{A}_T + \int (e^G - 1) \, dA_f.$$

Choose successively truncated quadratic velocity tests. Conservation of mass and energy controls A_f on compact velocity sets, while the B2 exponential

preparation estimate controls its velocity tails. Thus the contact currents are uniformly tight. The weak balance identity yields

$$\left| \int \phi_j(f_t - f_s) \right| \leq |t - s| \|v \cdot \nabla_x \phi_j\|_{L_2^\infty} \sup_r \int (1 + |v|^2) f_r + \int_{(s,t]} |\Delta \phi_j| d\Gamma.$$

Uniform integrability of the contact measure makes the right side a common modulus in probability and, on an action sublevel, after discarding an arbitrarily small contact-time set. Reparametrize those finitely many jumps by their cumulative contact mass; the density path is then equicontinuous in d_* . Helly–Arzelà compactness and a diagonal argument give a convergent subsequence, and the B2 closed Green graph identifies its limit as admissible.

On compact velocity sets the density coordinates converge strongly after velocity averaging of the balance equation. Hence $f_n f_{n,*} B dt dx dv dv_* d\omega$ converges weakly to A_f there; the tail estimate removes the truncation. The variational representation

$$\int \ell(d\Gamma/dA_f) dA_f = \sup_G \left\{ \int G d\Gamma - \int (e^G - 1) dA_f \right\}$$

then gives lower semicontinuity. Continuity of the terminal coordinates follows from the chosen uniform density topology. $\square$

1.2. State-dependent transfer without signed corrections. Let (f, qA_f) be a regular controlled path on $[0, T]$ and let g_0 be near f_0 in d_* on the same energy shell. Define g as the unique mild solution of

$$\partial_t g + v \cdot \nabla_x g = \mathcal{Q}_{qA_g}, \quad g(0) = g_0.$$

Lemma 1.2 (Positive multiplier transfer). *On every compact regular control chart there are $C, \delta > 0$ such that, if $d_*(f_0, g_0) < \delta$, the path (g, qA_g) exists on $[0, T]$, is positive and exactly balanced, and satisfies*

$$\sup_{t \leq T} d_*(f_t, g_t) \leq C d_*(f_0, g_0), \quad |\mathcal{A}_T(g, qA_g) - \mathcal{A}_T(f, qA_f)| \leq C d_*(f_0, g_0).$$

No signed collision-current correction is added.

Proof. For bounded q on the regular chart, the controlled Boltzmann collision map is locally Lipschitz in the weighted L^1 norm and preserves positivity and all collision invariants. Picard iteration of the mild equation therefore gives the unique positive path. Subtract the two mild equations, use the bilinear estimate

$$\|\mathcal{Q}_{qA_g} - \mathcal{Q}_{qA_f}\|_{L_2^1} \leq C(\|f\|_{L_2^1} + \|g\|_{L_2^1})\|f - g\|_{L_2^1},$$

and apply Gronwall. Approximation of the metric tests by the weighted L^1 class gives the d_* bound. Since the relative multiplier is the same q , the action difference is $\int \ell(q)(dA_g - dA_f)$ and is controlled by the same bilinear estimate. The balance equation holds by construction, so there is no residual and no appeal to a finite-dimensional correction of an infinite-dimensional PDE. $\square$

1.3. The Nisio semigroup. For $h \in \text{BUC}_{\mathfrak{w}}(\mathcal{E})$ define

$$(S_T h)(f_0) = \sup_{(f, \Gamma): f(0)=f_0} \{h(f_T) - \mathcal{A}_T(f, \Gamma)\}.$$

The zero-cost choice is the ordinary Boltzmann path $q = 1$.

Theorem 1.3 (Weighted strongly continuous Nisio semigroup). *The operators $(S_T)_{T \geq 0}$ form an order-preserving, constant-additive contraction semigroup on $\text{BUC}_{\mathfrak{w}}(\mathcal{E})$. On each energy shell,*

$$\sup_{f \in \mathcal{E}_M} |S_T h(f) - h(f)| \longrightarrow 0 \quad (T \downarrow 0).$$

The supremum is attained for bounded terminal rewards on an energy shell.

Proof. Concatenating admissible paths and using additivity of the action gives $S_{t+s} = S_t S_s$. Order preservation and constant additivity are immediate; taking the same path for two rewards gives the contraction estimate. For short times, the zero-cost path moves each test coordinate by $O_M(T)$. Conversely, an optimizer moving a coordinate by more than η must spend a positive amount determined by the entropy inequality and the balance formula. Uniform continuity of h on $\mathcal{E}_M$ therefore gives the stated uniform short-time limit. The weight is propagated by mass and energy conservation.

For an attaining sequence, truncate to a common action sublevel using the bounded reward. Theorem 1.1 supplies a convergent subsequence; continuity of the terminal reward and lower semicontinuity of the action make its limit optimal. $\square$

1.4. The correct nonlinear discounted resolvent. For $\lambda > 0$ define

$$(J_\lambda h)(f_0) = \sup_{(f, \Gamma): f(0)=f_0} \int_0^\infty e^{-t/\lambda} \left[\frac{1}{\lambda} h(f_t) - \ell(q_t) A_{f_t}(\mathbf{1}) \right] dt,$$

where the action density notation means integration over the collision variables. The exponential moment and energy shell bounds make the integral finite.

Theorem 1.4 (Nonlinear resolvent identity and accretive graph). *The maps J_λ are contractions and satisfy, for $0 < \mu < \lambda$,*

$$J_\lambda h = J_\mu \left(\frac{\mu}{\lambda} h + \left(1 - \frac{\mu}{\lambda} \right) J_\lambda h \right).$$

Define the graph $\mathcal{H}$ by

$$(u, g) \in \mathcal{H} \iff u = J_\lambda(u + \lambda g)$$

for one, and hence every, $\lambda > 0$. Then $\mathcal{H}$ is dissipative, $I - \lambda \mathcal{H}$ has full range for every $\lambda > 0$, and the semigroup generated by $\mathcal{H}$ is exactly (S_t) .

Proof. Split the discounted integral at a stopping time s and use the dynamic programming principle. Equivalently,

$$J_\lambda h = \sup_{s > 0} \left\{ \int_0^s e^{-t/\lambda} \left(\frac{h(f_t)}{\lambda} - L_t \right) dt + e^{-s/\lambda} J_\lambda h(f_s) \right\}.$$

Multiply this identity by the exponential density of a mean- μ waiting time and integrate in s . Fubini's theorem and the memoryless property split the mean- λ horizon into a mean- μ first stage followed by the continuation value. The resulting reward is exactly $(\mu/\lambda)h + (1 - \mu/\lambda)J_\lambda h$, proving the displayed nonlinear resolvent identity. No subtraction or linear composition identity is used.

The contraction follows by comparing two rewards under the same controls. The nonlinear identity proves that the graph definition is independent of λ . If $u_i = J_\lambda(u_i + \lambda g_i)$, contraction gives

$$\|u_1 - u_2\| \leq \|u_1 - u_2 + \lambda(g_1 - g_2)\|,$$

which is dissipativity. For every h , $u = J_\lambda h$ gives $(u, (h - u)/\lambda) \in \mathcal{H}$, so the range is full. The nonlinear Crandall–Liggett exponential formula defines the unique contraction semigroup. Applying the resolvent DPP repeatedly and letting the mesh tend to zero identifies that formula with the finite-horizon Nisio DPP, hence with S_t . $\square$

1.5. Exact law–hierarchy realization. In a bounded finite-volume hard-sphere domain, the exclusion condition gives a finite maximal particle number in every compact position cell. For a point process law P , let $\rho^{(k)}$ be its factorial moment densities.

Theorem 1.5 (Finite-volume Janossy reconstruction and BBGKY domain). *A sequence $(\rho^{(k)})_{k \geq 0}$ is the factorial hierarchy of a unique hard-sphere law with exponential number moment if and only if it is symmetric, consistent, supported on the exclusion configurations, Lenard positive, and satisfies $\sum_k z^k \|\rho^{(k)}\|_1/k! < \infty$ for some $z > 1$. The inverse is*

$$j_n = \frac{1}{n!} \sum_{k \geq 0} \frac{(-1)^k}{k!} \int \rho^{(n+k)} dX_{n+1} \cdots dX_{n+k},$$

and $P = \sum_n j_n dX_n$. On the Green–specular graph domain of B2, the Liouville evolution is intertwined exactly with the BBGKY hierarchy and preserves the realizable cone.

Proof. The exponential bound makes the inclusion–exclusion series for j_n absolutely convergent. Lenard positivity makes every Janossy functional nonnegative; consistency gives normalization. Conversely, direct factorial expansion of a point-process law yields the displayed conditions and inclusion–exclusion inversion. Absolute convergence proves uniqueness.

Apply the finite-particle Green identity to a smooth factorial observable. Bulk transport gives the free hierarchy term and the incoming/outgoing trace relation gives the binary BBGKY boundary term. The B2 closed trace theorem permits graph closure. Because the finite evolution is a push-forward of a probability law, its hierarchy remains Lenard positive and consistent; the exponential number bound is unchanged. $\square$

1.6. A quantitative microscopic graph core. Let F be a smooth cylinder in the limiting kinetic state. Truncate the B2 connected expansion at genealogy size K . For each retained genealogy solve backward the finite terminal-value transport equation with the exact specular boundary relation; call the sum $F_{\varepsilon,K}$.

Lemma 1.6 (Diagonal choice with nonuniform genealogy exponents). *There is a deterministic $K(\varepsilon) \uparrow \infty$ such that*

$$\sum_{|G| \leq K(\varepsilon)} C_G \varepsilon^{\alpha_G} \rightarrow 0, \quad \sum_{|G| > K(\varepsilon)} (CT)^{|G|} \rightarrow 0.$$

Proof. For every fixed K , the first sum tends to zero because it contains finitely many positive exponents. Choose a decreasing sequence ε_K so that the first sum is below 2^{-K} whenever $\varepsilon < \varepsilon_K$, and define $K(\varepsilon)$ as the largest K with that property. Reduce the short slicing interval so that $CT < 1$; then the geometric tail tends to zero. This diagonal argument requires no uniform lower bound on the genealogy-dependent exponents. $\square$

Theorem 1.7 (Realizable Liouville graph-core corrector). *The observables $F_\varepsilon = F_{\varepsilon,K(\varepsilon)}$ belong to the exact finite-particle Liouville graph domain, are factorial observables of realizable laws, and satisfy on every kinetic action sublevel*

$$F_\varepsilon \rightarrow F, \quad H_\varepsilon F_\varepsilon \rightarrow HF$$

locally uniformly, where H_ε is the microscopic nonlinear Hamiltonian and H the B2 kinetic Hamiltonian.

Proof. The backward terminal-value equation is solved separately on every finite-particle collision chamber. Its incoming boundary value is defined as the outgoing value after the exact specular reflection, so $F_{\varepsilon,K}$ lies in the graph domain without later regularization. Symmetrizing over labels makes it a factorial observable; Theorem 1.5 identifies it on the realizable cone.

For genealogies of size at most K , the B2 precontact theorem bounds every surplus contribution by $C_G \varepsilon^{\alpha_G}$. The normalized connected-tree estimate gives the geometric tail beyond K . Lemma 1.6 makes both errors vanish. Compact action sublevels provide uniform velocity/contact truncation, so the convergence is locally uniform. Differentiating the finite terminal equation gives the generator convergence with the same estimates. $\square$

1.7. Nonlinear semigroup convergence. Let S_t^ε be the finite-volume logarithmic semigroup on law observables transported to factorial hierarchies, and let S_t be Theorem 1.3.

Theorem 1.8 (Nonlinear Trotter–Kato limit). *For every weighted uniformly continuous kinetic observable h and every sequence of microscopic laws whose empirical states converge to f with uniform energy/exponential moments,*

$$S_t^\varepsilon h_\varepsilon \longrightarrow S_t h(f)$$

locally uniformly for t in compact intervals and f in kinetic action sublevels. The limiting semigroup is the unique one generated by the graph $\mathcal{H}$ of Theorem 1.4.

Proof. Exponential tightness and compact containment follow from B2 and Theorem 1.1. Apply Theorem 1.7 to a graph core of smooth cylinders. Its locally uniform generator convergence supplies both upper and lower extended generator inclusions by adding and subtracting a vanishing constant. Hence upper and lower half-relaxed limits are respectively a viscosity subsolution and supersolution of the resolvent equation for $\mathcal{H}$.

The contraction resolvent of Theorem 1.4 gives comparison: apply dissipativity to the doubled variables and then let the penalty vanish. Therefore the two relaxed limits coincide. Density of the core, contraction of both semigroups, and the exponential formula extend the convergence to every $h \in \text{BUC}_{\text{w}}$. Initial convergence and compact containment identify the value at $t = 0$, completing the nonlinear Trotter–Kato argument. $\square$

Theorem 1.9 (Round-twenty-one B4 closure). *The kinetic value theory has compact action sublevels, positive state-dependent control transfer, a weighted strongly continuous Nisio semigroup, the correct nonlinear resolvent identity and maximal dissipative graph, an exact realizable hierarchy intertwiner, a quantitative specular-domain graph core, and nonlinear semigroup convergence.*

Proof. Combine Theorems 1.1, 1.3, 1.4, 1.5, 1.7, and 1.8 with Lemma 1.2. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Twenty-One dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: qiqian@pku.edu.cn